

Certified Geometry-Aware Control Barrier Functions for Non-Convex Robots: From Planar Configuration-Space Fields to Articulated Manipulators

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Abstract—Geometry-aware control barrier functions (CBFs) that evaluate signed distance fields (SDFs) at closest boundary points need convex level sets, treat non-convex geometry through rotation-invariant bounding circles, and rely on a heuristic enclosure margin. We reduce collision avoidance between two non-convex rigid bodies to a condition on their boundaries at first contact and give two certified constructions of smooth barriers for it, neither of which computes closest points. For planar bodies, one smooth B-spline field per pair tabulates the configuration-space distance over the full relative pose and yields a single scalar barrier; a Bernstein branch-and-bound certifies its level for all continuous orientations without Minkowski sums. For spatial bodies and articulated robots, where such tables are intractable, the continuous level set of each body’s SDF is covered offline by boxes selected from Bernstein coefficients, and a certified curvature bound turns every box corner into a smooth barrier that is evaluated online without optimization. In both cases, certified levels enclose the true geometry, and the guarantees hold in continuous time and exact arithmetic. In simulation, the planar barrier prevented all collisions of non-convex rotating robots, unlike a closest-point barrier, and enabled docking that bounding circles preclude; a Franka Panda whose eight mesh links were each constrained against three non-convex obstacles avoided collision in all 30 trials in which the unfiltered motion collides, whereas a closest-point barrier collided in 7 and sphere and surface-sample barriers without margins in 17 and 19; two Pandas sharing a workspace kept their meshes apart.

Index Terms—Collision avoidance, control barrier functions, signed distance fields, configuration space, Bernstein polynomials, B-splines, multi-robot systems, manipulators.

I. Introduction

CONTROL barrier functions (CBFs) [1] turn safety conditions into constraints on the control input, typically enforced by a quadratic program (QP) that minimally modifies a nominal command. For collision avoidance, their value is determined by the geometric surrogate behind the barrier. Spheres and super-ellipsoids [2], [3] are smooth but conservative for irregular shapes; polytopic formulations [4], [5] are exact for convex pieces but become pairwise and piecewise for non-convex bodies; learned, point-cloud, and sampled barriers [6]–[9] trade exactness for generality.

Jo et al. [10] proposed geometry-aware CBFs on Bernstein-polynomial SDFs [11]: the barrier is the robot’s SDF evaluated at the closest point of the obstacle boundary, obtained from the Karush–Kuhn–Tucker (KKT)

conditions. That formulation has three limitations. L1) Its time derivative requires the closest point to move smoothly, which holds only for convex level sets. L2) For composite or non-convex geometry, the closest point jumps and the barrier is non-smooth; the remedy of [10], inflating obstacles by the robot’s bounding circle, ignores orientation. L3) The enclosure of the true geometry by the fitted level set rests on a user-chosen margin.

This article removes all three. Its starting point is that two compact bodies can only begin to intersect at a point that lies on both boundaries (Lemma 2). Keeping the SDF of one body above its level on the whole boundary of the other therefore suffices, and no closest point is needed. We certify this condition in two ways, according to the dimension of the relative pose. For planar bodies, the relative pose is three-dimensional, and the minimum is tabulated offline as one smooth field over (x, y, θ) per pair (Fig. 1); a single scalar barrier then encodes non-convex shapes and rotation. For spatial bodies and articulated robots, the relative pose has six or more dimensions and such tables are intractable, so the minimum is bounded online: the boundary is covered by small boxes and each box corner yields one smooth barrier.

A. Contributions

- 1) Boundary reduction and certified regions (Sec. IV): a first-contact lemma for compact bodies in $\mathbb{R}^n$, and certified levels of fitted SDFs that enclose the true geometry, for spline boundaries and for triangle meshes (resolves L3).
- 2) Tabulated barrier for planar pairs (Sec. V): one smooth barrier per pair of non-convex rotating bodies over $SE(2)$, whose level is certified on the explicit three-parameter first-contact set by a Bernstein branch-and-bound, together with certificates of its activation band and regularity (resolves L1 and L2).
- 3) Surface-cover barriers for spatial and articulated bodies (Sec. VI): a Bernstein cover of the continuous level set of each body’s SDF and a certified Taylor bound with a C^2 curvature majorant, which yield smooth barriers affine in the joint velocities, evaluated without closest points or online optimization.

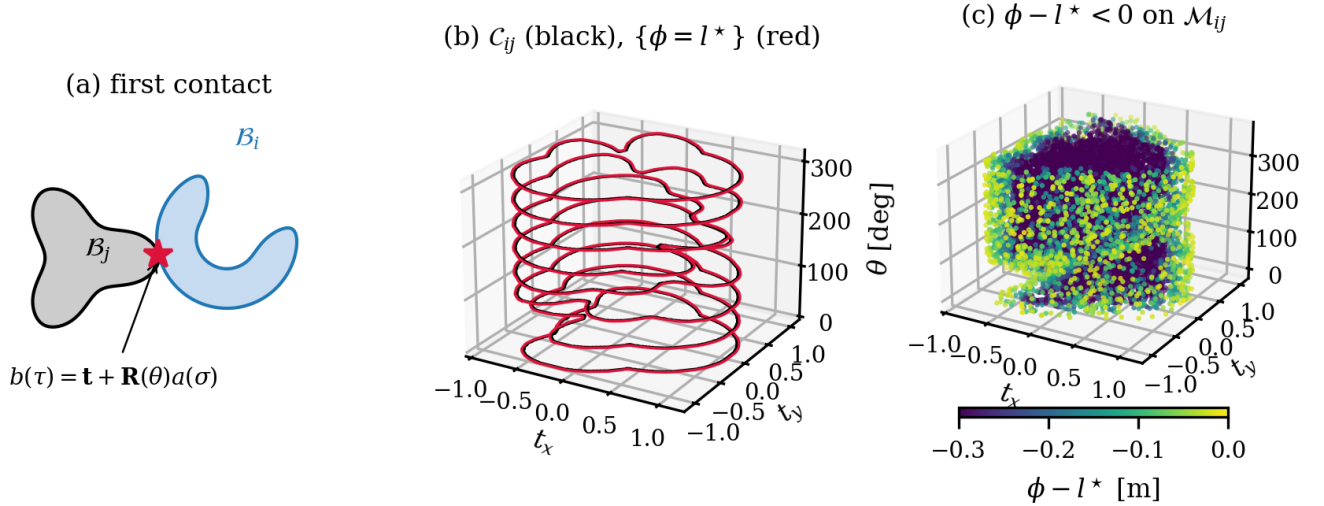


Fig. 1. Tabulated construction for planar bodies. (a) Every first contact occurs at a configuration $\mathbf{t} = b(\tau) - \mathbf{R}(\theta)a(\sigma)$ that joins boundary points of both bodies (Lemma 2). (b) The configuration-space obstacle C_{ij} of a C-shaped and a three-lobed body (black: rasterized reference, eight orientations) and the certified level set $\{\phi_{ij} = l_{ij}^*\}$ of one B-spline field (red). (c) The first-contact set $\mathcal{M}_{ij}$; the branch-and-bound of Sec. V-B certifies $\phi_{ij} < l_{ij}^*$ on all of it, for all continuous orientations.

- 4) Evaluation (Sec. VIII): planar docking, multi-robot, and static-obstacle experiments under single-integrator and unicycle dynamics against closest-point, bounding-circle, sampling-based, and ball-world baselines, a comparison of Bernstein-form representations and of both constructions on the same docking task, a 7-DOF Franka Panda among non-convex obstacles against sphere and surface-sample barriers, and two Pandas in a shared workspace.

B. Relation to Prior Work

This article keeps the problem formulation, the Bernstein-form distance fields, the pairwise multi-robot QP, and the experimental protocol of [10]. It replaces the closest-point barrier (retained as a baseline), the bounding-circle configuration space, and the heuristic margin. New are the first-contact reduction, both barrier constructions with their certificates and derivatives, the feasibility result, the representation analysis, and all experiments. The certified enclosure of planar boundaries by B-spline curves and the conversion of B-splines to Bernstein form follow [12] and are not claimed as contributions.

II. Related Work

Geometric barriers. Pairwise barriers between discs or spheres are the standard multi-robot certificates [2], and super-ellipsoids extend them to elongated vehicles [3]; unions of primitives multiply the constraints. For convex polytopes, signed distances through convex programs in the Minkowski-difference space yield exact barriers [4], [13], and smooth barriers exist between strongly convex regions [5]; non-convex bodies must be decomposed. Explicit smooth barriers for polygons compose edge-wise

bounds with smooth extrema [14]. Sampled distance functions yield nonsmooth barriers with quantified sampling error [9], and diffeomorphisms to ball worlds handle non-convex unsafe sets without undesired equilibria [15]; we compare with adaptations of both.

Distance fields for manipulators. Bernstein-polynomial distance fields of articulated robots support whole-body collision avoidance, with obstacle points queried against the robot field [11]; neural configuration-space distance functions serve as CBFs with robust margins [16], and differentiable distance fields map obstacle motion into joint space [17]. These works establish distance fields as barriers; they do not certify that the constraint holds on the whole continuous surface of the fitted or true geometry, which is what our surface covers provide.

Configuration-space obstacles and Bernstein bounds. Configuration-space obstacles of polygons can be constructed exactly [18], [19] or by grid convolution [20]; we use the latter only to produce training data. The convex-hull property and de Casteljau subdivision give rigorous, successively tighter bounds on polynomials over boxes [21], [22]; they were used in [12] to certify planar spline enclosures, and we use them in up to three variables on tensor-product fields.

III. Preliminaries and Problem Statement

CBFs. Consider $\dot{\mathbf{z}} = f(\mathbf{z}) + g(\mathbf{z})\mathbf{u}$ with f, g locally Lipschitz, $\mathbf{u} \in \mathcal{U}$, and a C^1 function h with $\mathcal{H} = \{\mathbf{z} \mid h(\mathbf{z}) \geq 0\}$.

Lemma 1 ([1]). If a locally Lipschitz feedback $\mathbf{k}(\mathbf{z}) \in \mathcal{U}$ satisfies $L_f h + L_g h \mathbf{k} \geq -\gamma h$ with $\gamma > 0$ on an open set containing $\mathcal{H}$, then $\mathcal{H}$ is forward invariant.

Bernstein form. The cubic Bernstein basis $B_r(\xi) = \binom{3}{r} \xi^r (1 - \xi)^{3-r}$ is nonnegative and sums to one, so

$p = \sum_r \beta_r B_r$ satisfies $\min_r \beta_r \leq p \leq \max_r \beta_r$ on $[0, 1]$ (convex-hull property), also for tensor products. With M_B mapping Bernstein to power coefficients and $T_{jr} = \binom{r}{j} \xi_0^{r-j} \Delta^j$ ($j \leq r$), the coefficients of p restricted to $[\xi_0, \xi_0 + \Delta]$ and reparametrized to $[0, 1]$ are

$$\beta' = S(\xi_0, \xi_0 + \Delta) \beta, \quad S = M_B^{-1} T(\xi_0, \Delta) M_B, \quad (1)$$

applied along each axis; they converge to the values of p as $\Delta \rightarrow 0$. On each span, a uniform cubic B-spline with control values $\mathbf{P}_c, \dots, \mathbf{P}_{c+3}$ has Bernstein coefficients

$$\beta_c = Q [\mathbf{P}_c, \dots, \mathbf{P}_{c+3}]^\top, \quad Q = \frac{1}{6} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 4 & 1 \end{bmatrix}, \quad (2)$$

so bounds on a tensor-product spline, and on its derivatives, follow from finitely many coefficients per cell [12], [21].

Problem. Robot bodies and obstacles are compact sets $\mathcal{G}_i \subset \mathbb{R}^n$, $n \in \{2, 3\}$ (the ground truth, not necessarily convex), rigidly attached to frames with poses $(\mathbf{R}_i, \mathbf{p}_i)$ that depend on the state: the pose of a mobile robot, or the forward kinematics of a link. Given nominal inputs, find minimally modified inputs under which no two bodies ever intersect, with constraints whose correctness is certified rather than sampled.

IV. Certified Regions and First Contact

A. First Contact

Lemma 2 (First contact). Let $\mathcal{K}_A(s), \mathcal{K}_B(s) \subset \mathbb{R}^n$, $s \in [0, T]$, be images of compact sets under rigid motions that are continuous in s , disjoint at $s = 0$ and intersecting at $s = T$. Then at $s^* = \inf\{s \mid \mathcal{K}_A(s) \cap \mathcal{K}_B(s) \neq \emptyset\} > 0$, some point lies on both boundaries, $\partial\mathcal{K}_A(s^*) \cap \partial\mathcal{K}_B(s^*) \neq \emptyset$.

Proof. The set of intersecting parameters is closed, so the bodies intersect at $s^* > 0$; pick $\mathbf{z}$ in both. If $\mathbf{z}$ were interior to $\mathcal{K}_B(s^*)$, the point of $\mathcal{K}_A$ that is at $\mathbf{z}$ at s^* would, by continuity, lie in $\mathcal{K}_B(s)$ for $s < s^*$ close to s^* , a contradiction. Symmetrically, $\mathbf{z}$ is not interior to $\mathcal{K}_A(s^*)$, since the point of $\mathcal{K}_B$ at $\mathbf{z}$ would lie in $\mathcal{K}_A$ before s^* . $\square$

The lemma requires neither convexity nor smoothness. It reduces collision avoidance to excluding coincident boundary points, provided that the trajectory starts collision-free.

B. Certified Regions

Each body is replaced by a region $\mathcal{B}_i \supseteq \mathcal{G}_i$ described by a fitted smooth function whose enclosure of $\mathcal{G}_i$ is certified; safety for the regions then implies safety for the ground truth.

Planar spline boundaries [12]. The boundary is a closed periodic cubic B-spline $a_i : \mathbb{S}^1 \rightarrow \mathbb{R}^2$, fitted to outward-offset samples of $\mathcal{G}_i$ and accepted only if every Bézier control hull (after subdivision) misses $\mathcal{G}_i$ and the curve winds around a point of $\mathcal{G}_i$. The region $\mathcal{B}_i$ is the curve with the points of nonzero winding number; for connected $\mathcal{G}_i$, $\mathcal{G}_i \subseteq \mathcal{B}_i$ and $\partial\mathcal{B}_i \subseteq a_i(\mathbb{S}^1)$.

Level sets of SDFs. Let ϕ_i be a tensor-product cubic B-spline fitted to (approximate) signed distances of $\mathcal{G}_i$ on a box Ω_i , and let $\kappa_i(\mathbf{g})$ be a certified bound on $\|\nabla^2 \phi_i\|$ over the ball of radius δ around $\mathbf{g}$ (Sec. VI; valid for δ up to the cell size). Let $\mathcal{P}$ be a grid of spacing s that covers $\mathcal{G}_i$, $\delta = s\sqrt{n}/2$, and $\mathcal{P}_\delta$ its points within distance δ of $\partial\mathcal{G}_i$, computed with exact point-to-boundary distances (for a triangle mesh, point-to-triangle distances).

Proposition 1. With $l_i = \max_{\mathbf{g} \in \mathcal{P}_\delta} (\phi_i(\mathbf{g}) + \|\nabla \phi_i(\mathbf{g})\| \delta + \frac{1}{2} \kappa_i(\mathbf{g}) \delta^2) + \epsilon$, $\epsilon > 0$, the boundary satisfies $\partial\mathcal{G}_i \subset \{\phi_i < l_i\}$.

Proof. Every $\mathbf{x} \in \partial\mathcal{G}_i$ has a grid point $\mathbf{g}$ with $\|\mathbf{x} - \mathbf{g}\| \leq \delta$, which lies in $\mathcal{P}_\delta$; Taylor's theorem bounds $\phi_i(\mathbf{x})$ by the bracket. $\square$

The fit affects only the size of l_i , never its validity, so the training targets may be approximate. We write $\mathcal{S}_i = \{\phi_i = l_i\}$ for the certified surface.

V. Tabulated Barrier for Planar Pairs

A. Configuration-Space Field and Barrier

For planar bodies with poses $(\mathbf{p}_i, \theta_i)$, the configuration of robot i relative to robot j is $\mathbf{q}_{ij} = (\mathbf{t}_{ij}, \theta_{ij})$ with $\mathbf{t}_{ij} = \mathbf{R}(-\theta_j)(\mathbf{p}_i - \mathbf{p}_j)$ and $\theta_{ij} = \theta_i - \theta_j$, and the robots intersect if and only if $\mathbf{q}_{ij}$ lies in the configuration-space obstacle $\mathcal{C}_{ij} = \{(\mathbf{t}, \theta) \mid (\mathbf{R}(\theta)\mathcal{B}_i + \mathbf{t}) \cap \mathcal{B}_j \neq \emptyset\}$. We fit, offline, a field on $\Omega_{ij} = [-D_{ij}, D_{ij}]^2 \times \mathbb{S}^1$ that approximates the signed distance to $\mathcal{C}_{ij}$ (negative inside),

$$\phi_{ij}(\mathbf{t}, \theta) = \sum_{a,b,c} W_{abc} N_a(t_x) N_b(t_y) \tilde{N}_c(\theta), \quad (3)$$

with uniform cubic B-spline bases and a periodic basis $\tilde{N}_c$ in θ , and $D_{ij} > \rho_i + \rho_j$, where ρ_i bounds the radius of $\mathcal{B}_i$. Given a certified level l_{ij}^* (Sec. V-B), the barrier is

$$h_{ij} = \phi_{ij}(\mathbf{q}_{ij}) - l_{ij}^*, \quad (4)$$

which is C^2 in the poses. As training data, we rasterize $\mathcal{C}_{ij}$ at 96 orientations by convolving occupancy masks, take Euclidean distance transforms, and fit W by separable ridge regression; the certificate never samples orientations.

Proposition 2. With $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{g}_t = \nabla_{\mathbf{t}} \phi_{ij}$, $g_\theta = \partial_\theta \phi_{ij}$, and $SE(2)$ single integrators $\dot{\mathbf{p}}_i = \mathbf{v}_i$, $\dot{\theta}_i = \omega_i$,

$$\dot{h}_{ij} = (\mathbf{R}(\theta_j) \mathbf{g}_t)^\top (\mathbf{v}_i - \mathbf{v}_j) + g_\theta \omega_i - (\mathbf{g}_t^\top J \mathbf{t}_{ij} + g_\theta) \omega_j. \quad (5)$$

Proof. $\dot{\mathbf{t}}_{ij} = -\omega_j J \mathbf{t}_{ij} + \mathbf{R}(-\theta_j)(\mathbf{v}_i - \mathbf{v}_j)$ and $\dot{\theta}_{ij} = \omega_i - \omega_j$; apply the chain rule. $\square$

For unicycles, $\mathbf{v}_k = v_k \mathbf{e}(\theta_k)$ with $\mathbf{e}(\theta) = [\cos \theta, \sin \theta]^\top$ turns (5) into $\dot{h}_{ij} = a_i v_i + g_\theta \omega_i - a_j v_j - (\mathbf{g}_t^\top J \mathbf{t}_{ij} + g_\theta) \omega_j$ with $a_k = (\mathbf{R}(\theta_j) \mathbf{g}_t)^\top \mathbf{e}(\theta_k)$, still affine in the inputs; we monitor the norm of its coefficient vector in the experiments.

For a robot among static obstacles, the configuration-space obstacle of a union is the union of the configuration-space obstacles, so one field over the workspace and θ encodes all obstacles in a single barrier whose online cost is independent of their number.

Algorithm 1 Contact certificate for a pair (i, j)

Require: spans of a and b , field ϕ_{ij} , tolerance $\varepsilon > 0$

- 1: $\mathcal{Q} \leftarrow$ all span pairs $\times n_\theta$ angle intervals; $\ell \leftarrow \max_{\mathcal{Q}} \text{LB}$
- 2: while $\mathcal{Q} \neq \emptyset$ do
- 3: remove every $\mathcal{R}$ with $\text{UB}(\mathcal{R}) < \ell + \varepsilon$
- 4: bisect each remaining $\mathcal{R}$ along its widest parameter image
- 5: $\ell \leftarrow \max(\ell, \max_{\mathcal{Q}} \text{LB})$
- 6: end while
- 7: return $l_{ij}^* = \ell + \varepsilon$

B. Contact Certificate

By Lemma 2 applied to $\mathcal{B}_i$ and $\mathcal{B}_j$ with $\partial\mathcal{B} \subseteq a(\mathbb{S}^1)$, every first contact occurs in

$$\mathcal{M}_{ij} = \{(b(\tau) - \mathbf{R}(\theta)a(\sigma), \theta) \mid \sigma, \tau, \theta \in \mathbb{S}^1\}, \quad (6)$$

an explicit three-parameter family; the configuration-space obstacle, which would require a Minkowski sum for every orientation, is never formed.

Definition 1. A level l_{ij}^* is a contact certificate if $\mathcal{M}_{ij} \subset \Omega_{ij}$ and $\phi_{ij} < l_{ij}^*$ on $\mathcal{M}_{ij}$.

Theorem 1. Let l_{ij}^* be a contact certificate, and let a continuous trajectory start with $\mathbf{q}_{ij}(0) \notin \mathcal{C}_{ij}$ and satisfy $h_{ij} \geq 0$ whenever $\mathbf{q}_{ij} \in \Omega_{ij}$. Then $\mathcal{B}_i$ and $\mathcal{B}_j$, and hence $\mathcal{G}_i$ and $\mathcal{G}_j$, never intersect.

Proof. A first contact would lie in $\mathcal{M}_{ij} \subset \Omega_{ij}$, where $h_{ij} < 0$. $\square$

Branch-and-bound. After (2), both boundaries are piecewise cubic Bézier curves. A region $\mathcal{R}$ is a span of a with a parameter interval, a span of b with an interval, and an angle interval. By (1), the sub-curves lie in the hulls of their restricted control points; each rotated control point of a traces an arc whose bounding box is exact, so $\mathbf{t}$ ranges over a box $[\underline{\mathbf{b}} - \bar{\mathbf{a}}, \bar{\mathbf{b}} - \underline{\mathbf{a}}]$ on the part of $\mathcal{M}_{ij}$ covered by $\mathcal{R}$. The maximum of the Bernstein coefficients of ϕ_{ij} restricted to this box and angle interval is an upper bound $\text{UB}(\mathcal{R})$ ($+\infty$ if the box leaves Ω_{ij}); the value at the midpoint parameters is attained on $\mathcal{M}_{ij}$ and is a lower bound $\text{LB}(\mathcal{R})$ of $\max_{\mathcal{M}_{ij}} \phi_{ij}$.

Proposition 3. If Algorithm 1 terminates, l_{ij}^* is a contact certificate and $l_{ij}^* - \max_{\mathcal{M}_{ij}} \phi_{ij} \leq \varepsilon$. It terminates for every $\varepsilon > 0$ whenever $\mathcal{M}_{ij}$ lies in the interior of Ω_{ij} .

Proof. Every point of $\mathcal{M}_{ij}$ lies in a region removed with $\text{UB} < \ell + \varepsilon \leq l_{ij}^*$ (ℓ is nondecreasing), and a finite UB requires the box to lie in Ω_{ij} ; ℓ is attained on $\mathcal{M}_{ij}$. For termination, the control-hull diameters of the sub-curves are bounded by the curve derivatives times the parameter lengths, so the image widths of any nested chain of surviving regions tend to zero; the restricted coefficients of a cubic cell differ from its values by at most a constant times the box diameter, so $\text{UB} - \text{LB} \rightarrow 0$ uniformly, and a region survives only while $\text{UB} - \text{LB} \geq \varepsilon$. $\square$

Termination also certifies $\mathcal{M}_{ij} \subset \Omega_{ij}$, so the radius bounds ρ_i affect only efficiency. Two further certificates use the same bounds. The activation band: $h_{ij} > 0$ wherever $D_{ij} - \delta \leq \|\mathbf{t}\|_\infty \leq D_{ij}$, so a pair enters the QP with a positive barrier; the minimum restricted coefficient of every cell meeting the band must exceed l_{ij}^* . Regularity: bisecting every cell whose coefficient range contains l_{ij}^* until either l_{ij}^* leaves the range or the coefficients of one partial derivative have one strict sign certifies $\nabla\phi_{ij} \neq 0$ on $\{h_{ij} = 0\}$, so the barrier row never vanishes there. All certificates assume exact arithmetic; our implementation uses double precision without outward rounding.

C. Representations

The global Bernstein polynomial of [10], piecewise Bézier patches, and cubic B-splines are all cellwise Bernstein forms, so Algorithm 1 applies to each. Patches that share only boundary coefficients are C^0 : their gradient jumps across cell faces, so the constraint normal jumps whenever the relative configuration crosses a face while the barrier is active, which violates the C^1 requirement of Lemma 1 and appears as chattering and persistent violations (Sec. VIII-C). The cubic B-spline is C^2 with about a third as many coefficients per cell; the global polynomial is smooth but, being one patch, cannot localize sharp non-convex features.

VI. Surface-Cover Barriers for Spatial and Articulated Bodies

For spatial rigid bodies, the relative pose is six-dimensional, and a tensor-product field over it would need about 6×10^7 coefficients at 20 per axis. We therefore keep one SDF per body and bound the first-contact condition online. Let body A (e.g., a link) have the certified surface $\mathcal{S}_A = \{\phi_A = l_A\}$ in its frame, and let body B (e.g., an obstacle) have the field ϕ_B and level l_B in its frame, both from Proposition 1. We write $\mathbf{x}$ for points of A mapped into B 's frame by the current relative pose.

A. Certified Cover of the Continuous Surface

On each cell of ϕ_A , the Bernstein coefficients of $\phi_A - l_A$ follow from (2). If they all have one strict sign on a box, the box contains no point of $\mathcal{S}_A$ by the convex-hull property. Starting from the cells and halving every box whose coefficients change sign, restricted by (1), until its side is at most $s_{\max}$, yields finitely many boxes $\{\mathcal{Q}_k\}$ with centers $\bar{\mathbf{c}}_k$ and half-diagonal r whose union contains all of $\mathcal{S}_A$, not a sample of it. The cover is computed once per body.

B. Curvature Majorant and Corner Barriers

Cellwise bounds on $\|\nabla^2\phi_B\|_F$ follow from the Bernstein coefficients of the second derivatives. They are piecewise constant, and a barrier that switches between them would be discontinuous. We therefore use a C^2 majorant: a cubic B-spline κ_B on the grid of ϕ_B whose control value with

index m is the maximum of the cellwise bounds over cells $m - 4, \dots, m + 1$ along each axis. Since cubic B-spline weights are nonnegative and sum to one, $\kappa_B(\mathbf{c})$ bounds the Hessian on every cell adjacent to the cell of $\mathbf{c}$, and hence on the ball of radius r around $\mathbf{c}$ whenever r does not exceed the cell size.

Proposition 4. Let a box $\mathcal{Q}$ of A have center $\mathbf{c}$, corners $\mathbf{v}_1, \dots, \mathbf{v}_8$, and half-diagonal r in B 's frame, and define for each corner

$$h_k = \phi_B(\mathbf{c}) + \nabla\phi_B(\mathbf{c})^\top(\mathbf{v}_k - \mathbf{c}) - \frac{1}{2}\kappa_B(\mathbf{c})r^2 - l_B. \quad (7)$$

Then $\min_{\mathbf{x} \in \mathcal{Q}} \phi_B(\mathbf{x}) - l_B \geq \min_k h_k$.

Proof. For $\mathbf{x} \in \mathcal{Q}$, $\|\mathbf{x} - \mathbf{c}\| \leq r$, and Taylor's theorem gives $\phi_B(\mathbf{x}) \geq \phi_B(\mathbf{c}) + \nabla\phi_B(\mathbf{c})^\top(\mathbf{x} - \mathbf{c}) - \frac{1}{2}\kappa_B(\mathbf{c})r^2$. The right-hand side is affine in $\mathbf{x}$, and the rigidly moved box is the convex hull of its corners, so its minimum is attained at a corner. $\square$

Each h_k is C^2 in the pose because ϕ_B and κ_B are. The bound is applied only to boxes whose ball lies in the domain of ϕ_B ; we choose the domain so that every box meeting $\{\phi_B \leq l_B\}$ has its ball inside it, so other boxes cannot meet this set. The bound is also tight up to the box size:

Proposition 5. Under the assumptions of Proposition 4, $0 \leq \min_{\mathbf{x} \in \mathcal{Q}} \phi_B(\mathbf{x}) - l_B - \min_k h_k \leq \|\nabla\phi_B(\mathbf{c})\|r + \frac{1}{2}\kappa_B(\mathbf{c})r^2$.

Proof. The lower inequality is Proposition 4. For the upper one, $\mathbf{c} \in \mathcal{Q}$ gives $\min_{\mathcal{Q}} \phi_B \leq \phi_B(\mathbf{c})$, and $\|\mathbf{v}_k - \mathbf{c}\| = r$ gives $h_k \geq \phi_B(\mathbf{c}) - \|\nabla\phi_B(\mathbf{c})\|r - \frac{1}{2}\kappa_B(\mathbf{c})r^2 - l_B$. $\square$

With $\|\nabla\phi_B\| \approx 1$, the barriers therefore stop body A at most about $l_A + l_B + r + \frac{1}{2}\kappa_B r^2$ from B beyond the fitting error, and this bound vanishes with the box size and the certified levels.

Theorem 2. Suppose $\{\phi_A \leq l_A\}$ and $\{\phi_B \leq l_B\}$ are compact subsets of their domains, the bodies are initially disjoint with $\{\phi_A \leq l_A\} \cap \{\phi_B < l_B\} = \emptyset$, and along a continuous trajectory $h_k \geq 0$ for every corner of every box of the cover whose ball lies in the domain of ϕ_B , and every box that meets $\{\phi_B \leq l_B\}$ is such a box. Then $\mathcal{G}_A$ and $\mathcal{G}_B$ never intersect.

Proof. By Proposition 4, $\phi_B \geq l_B$ on $\mathcal{S}_A$ at all times. For $\epsilon > 0$, if $\{\phi_A \leq l_A\}$ met $\{\phi_B \leq l_B - \epsilon\}$, Lemma 2 would give a first point on both boundaries, where $\phi_A = l_A$ and $\phi_B = l_B - \epsilon$, a contradiction. Hence $\{\phi_A \leq l_A\} \cap \{\phi_B < l_B\} = \emptyset$ throughout. A first contact of $\mathcal{G}_A$ and $\mathcal{G}_B$ would lie on $\partial\mathcal{G}_A \cap \partial\mathcal{G}_B$ (Lemma 2), where $\phi_A < l_A$ and $\phi_B < l_B$ by Proposition 1. $\square$

C. Derivative for Articulated Robots

Let link A move with joint positions $\mathbf{q}$ in front of a static body B , so that the box center and corner move as $\dot{\mathbf{c}} = J_{\mathbf{c}}(\mathbf{q})\dot{\mathbf{q}}$ and $\dot{\mathbf{v}}_k = J_{\mathbf{v}_k}(\mathbf{q})\dot{\mathbf{q}}$, with point Jacobians

whose column for a revolute joint j before the link is $\mathbf{z}_j \times (\cdot - \mathbf{o}_j)$. Differentiating (7) with $\mathbf{g} = \nabla\phi_B(\mathbf{c})$ gives

$$\dot{h}_k = (\mathbf{g} + \nabla^2\phi_B(\mathbf{c})(\mathbf{v}_k - \mathbf{c}) - \frac{r^2}{2}\nabla\kappa_B(\mathbf{c}))^\top J_{\mathbf{c}}\dot{\mathbf{q}} + \mathbf{g}^\top(J_{\mathbf{v}_k} - J_{\mathbf{c}})\dot{\mathbf{q}}, \quad (8)$$

which is affine in $\dot{\mathbf{q}}$; the same computation applies to two moving bodies in the frame of B . No closest point and no optimization are involved: online, ϕ_B , its first two derivatives, and κ_B are evaluated at box centers.

Pruning. Boxes whose certified lower bound exceeds an activation threshold $\eta > 0$ need no constraint: their corner barriers are at least η , and by continuity a box re-enters the QP with barriers of at least η (in discrete time, η must exceed the decrease per step). We group the boxes of each body into clusters with center $\bar{\mathbf{c}}$ and radius R (covering the boxes' balls) and skip a cluster if $\phi_B(\bar{\mathbf{c}}) - GR - \frac{1}{2}\bar{M}r^2 - l_B \geq \eta$, where G and $\bar{M}$ are certified bounds of $\|\nabla\phi_B\|$ and of κ_B on the cells that the cluster and the majorants' supports can reach; this implies $h_k \geq \eta$ for all its corners. Remaining boxes are pruned individually with $\phi_B(\mathbf{c}) - \|\mathbf{g}\|r - \frac{1}{2}\kappa_B(\mathbf{c})r^2 - l_B \geq \eta$.

VII. Safety Filter

Collecting the active barriers $\mathcal{A}$ of all pairs—one per active planar pair, one per active box corner otherwise—the filter is

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathcal{U}} \|\mathbf{u} - \mathbf{u}^{\text{nom}}\|^2 \quad \text{s.t.} \quad \dot{h}(\mathbf{u}) \geq -\gamma h \quad \forall h \in \mathcal{A}, \quad (9)$$

with $\mathbf{u}$ the stacked planar velocities or the joint velocities. With polytopic input bounds, (9) is a QP, which we solve exactly as a least-distance program by non-negative least squares [23], warm-started by constraint generation.

Proposition 6 (Feasibility on the safe set). For driftless dynamics—planar single integrators, unicycles, or velocity-controlled joints—and any $\mathcal{U} \ni \mathbf{0}$, $\mathbf{u} = \mathbf{0}$ is feasible for (9) whenever $h \geq 0$ for all $h \in \mathcal{A}$.

Proof. All barrier derivatives are linear in $\mathbf{u}$ without drift, so $\mathbf{u} = \mathbf{0}$ gives $\dot{h} = 0 \geq -\gamma h$. $\square$

Corollary 1. If all barriers are initially nonnegative, barriers enter $\mathcal{A}$ only with positive values (activation band or pruning threshold), and the minimizer of (9) is locally Lipschitz, then in continuous time no pair of bodies, and no pair of ground-truth shapes, ever intersects.

Proof. Lemma 1 keeps the active barriers nonnegative, Proposition 6 keeps the QP feasible, and Theorems 1 and 2 conclude. $\square$

Feasibility does not imply progress: exact non-convex geometry can hook and block bodies [24], and we do not resolve such deadlocks. In the experiments, a slack variable is added only if (9) is infeasible, which can happen only after a discrete-time violation.

VIII. Experiments

A. Planar Setup

Robot A is a C-shaped body and robot B a three-lobed body with smooth periodic B-spline boundaries ($\rho_A = 0.529$ m, $\rho_B = 0.579$ m), which also serve as ground truth. Unless stated otherwise, the pair field is a cubic B-spline with $45 \times 45 \times 48$ cells (1.1×10^5 coefficients), Algorithm 1 uses $n_\theta = 32$ and $\varepsilon = 2$ mm, and control uses $SE(2)$ single integrators with $\gamma = 5$, $v_{\max} = 1$ m/s, $\omega_{\max} = 2$ rad/s, a 10-ms step, and a proportional nominal controller. Computations run on an AMD Ryzen 9 3900X CPU with an NVIDIA RTX 4070 GPU (fitting and certification in double precision on the GPU; control in Python on the CPU). Ground-truth shapes are dense polygons; collisions are checked at every step and, between steps, along the interpolated poses with a rigid-motion bound and bisection down to 0.1 mm.

Baselines. The closest-point barrier is the signed distance between the fitted bodies, sampled as polygons, minus a margin $m = 2$ cm, with the derivative at one closest pair; it removes the fitting error of [10] but keeps its single-closest-point structure. The circle barrier [2] uses the minimum enclosing circle of each fitted body, with body-frame center, and the same margin.

B. Representations and Conservatism

Table I compares representations with about 1.1×10^5 coefficients. Both piecewise fields are certified, including band and regularity; the B-spline level is 14% lower than the C^0 level, and its gradient is continuous to numerical precision. The global polynomial of [10] is 2.8 times less accurate near the boundary, and its sampled maximum on $\mathcal{M}$ already exceeds the certified B-spline level, so no certificate for it could be less conservative.

The certified level bounds how much the field overestimates the distance at the obstacle boundary. Beyond moderate model sizes it is set by the training targets rather than by the representation: coarsening the raster from 1 to 2 cm doubles l^* (38.2 mm), refining it to 0.5 cm halves it (10.7 mm), whereas refining the B-spline cells from 8.3 to 4.7 cm leaves it at 18.6–18.7 mm. The tolerance trades millimeters for time (ε from 0.5 to 4 mm: l^* from 18.5 to 21.9 mm, time from 4.3 to 2.8 s). A per-cell certificate shows that only 3.5% of the 38,712 cells met by $\mathcal{M}$ have bounds above $l^*/2$, so a spatially varying level could reduce the conservatism considerably.

C. Docking into a Tapered Receptacle

A fixed module carries two panel arrays and, in its front face, a tapered receptacle; a vehicle has a hull, two nozzles, two panel arrays, and a 0.70-m nose that narrows from 0.32 to 0.10 m (Fig. 2). The receptacle is the seated nose offset outward by 32 mm everywhere, so the nose seats fully (0.668 m past the mouth) only if the barrier admits configurations within 32 mm of contact, and only within a narrow band of orientations. The certified boundaries need

TABLE I
Representations on the C-Shape/Three-Lobed Pair

	B-spline (C^2)	Bézier (C^0)	global $Q=48$	Bernstein $Q=23$ [10]
coefficients [10^3]	111	115	111	12
RMSE near boundary [mm]	2.6	2.6	6.1	9.6
max. error near bd. [mm]	17.0	21.2	48.4	79.8
sampled $\max_{\mathcal{M}} \phi$ [mm]	16.7	17.1	23.3	34.1
certified l^* [mm]	19.9	23.2	—	—
certification time [s]	3.6	2.0	—	—
max. gradient jump	9×10^{-6}	1.69	0	0

Errors: against the reference signed distance d at 16 held-out orientations, where $|d| < 0.1$ m. $\max_{\mathcal{M}} \phi$ is sampled at 2×10^5 points and lower-bounds any valid level; the global polynomials were not certified.

TABLE II
Docking with Different Barriers

	ours			closest	
	tabulated (B-spline)	tabulated (C^0 Bézier)	cover (5 mm)	point	circle
certified level [mm]	12.3	19.0	3.3, 3.3	—	—
nose depth [m]	0.666	0.692	0.690	0.671	−0.084
min. true gap [mm]	24.8	6.1	10.4	28.4	102.5
min. h [mm]	0.0	−15.8	0.0	−0.9	—
input variation	3.6	161	7.9	3.8	2.5
barriers, max.	1	1	1516	1	1
time per step [ms]	0.61	0.52	4.6	6.82	0.25

Nose depth: tip past the mouth at the end (full seating with a 32-mm gap: 0.668 m). Tabulated fields: $82^2 \times 384$ and 30^3 cells. Cover: levels of the station and craft fields, 1548 boxes. Time: median, complete filter. The circle barrier's h is in m^2 and stayed nonnegative.

offsets of 4 and 5 mm; training targets come from a 0.25-cm raster at 576 orientations; the field has $82 \times 82 \times 384$ cells (2.8×10^6 coefficients), and Algorithm 1 certifies $l^* = 12.3$ mm in 34 s. The vehicle starts 3.4 m away, rotated by 40° , and its nominal goal lies 10 cm past full seating.

Our barrier docks smoothly while respecting its constraint (Table II): the nose stops 2 mm short of full seating at $h = 0$, with a final ground-truth gap of 32 mm. Both C^0 fields push the nose deeper only by violating their constraints by 15.8 and 9.6 mm, with the constraint normal switching across cell faces and the input chattering (total variation 161 and 94.5, against 3.6). The closest-point barrier docks as smoothly here, because the contact stays on the tapered walls and the closest pair does not jump, but costs 11 times more per step; the circles stop the nose 8 cm in front of the mouth. Refining the step from 10 to 1 ms leaves our barrier nonnegative and the final pose unchanged, whereas the violation of the C^0 field stays at 15–16 mm while its input variation grows tenfold: the violation is caused by gradient jumps, not by discretization. With 96 instead of 384 angular cells, the level hardly changes (12.4 mm), but the nose stops 1.5 cm earlier: coarse angular cells round off the narrow ridge of free orientations near full seating.

Both constructions on one task. The surface-cover barriers of Sec. VI apply unchanged to planar bodies.

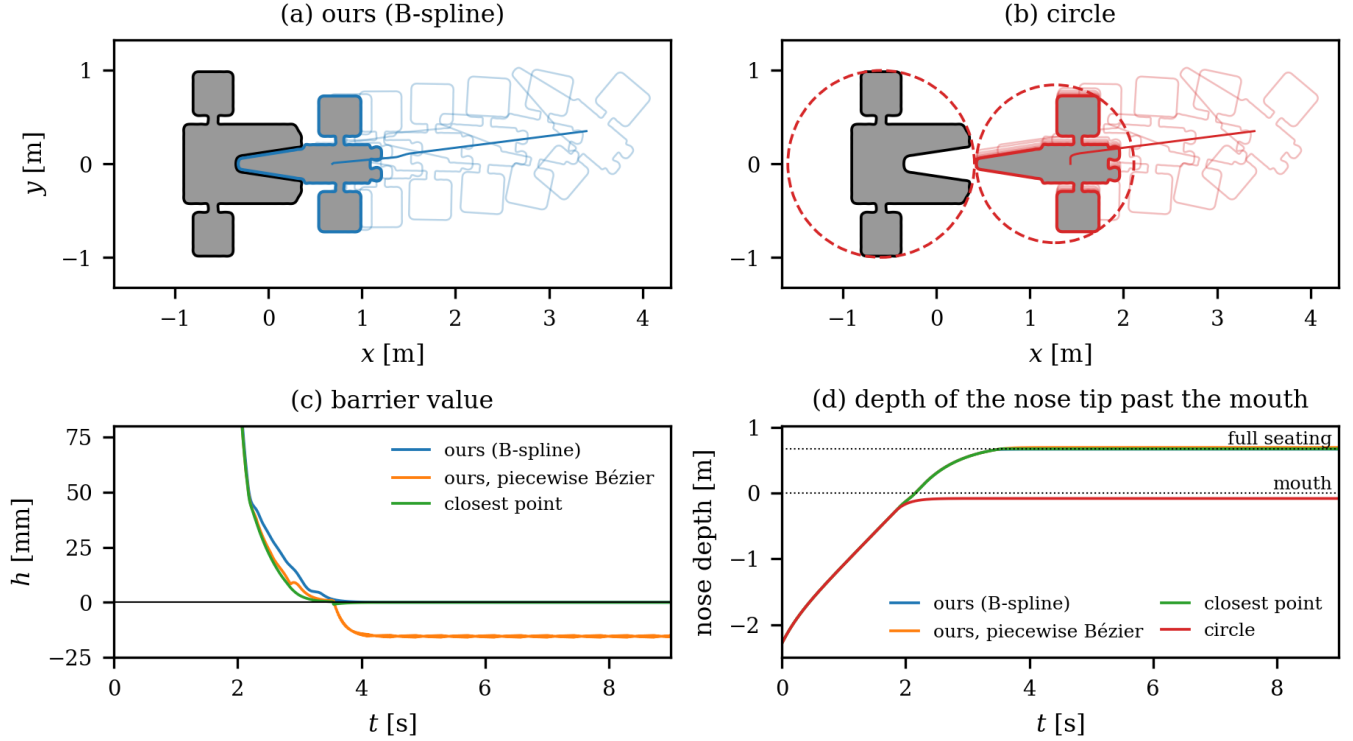


Fig. 2. Docking into a tapered receptacle with the module fixed. (a) Our barrier: the vehicle rotates while approaching and seats its 0.70-m nose to within 2 mm of full seating. (b) Minimum enclosing circles (dashed): the vehicle stops 8 cm in front of the mouth. Poses every 0.6 s; gray: ground truth; colored and black: certified fitted boundaries. (c) Barrier value: ours keeps $h \geq 0$, the C^0 field violates its constraint by 16 mm, and the closest-point barrier by 0.9 mm. (d) Depth of the nose tip past the mouth (full seating: 0.668 m).

With one workspace field per body (1-cm and 1.25-cm cells), levels certified against the exact polygon distances (3.3 mm each), and 1548 boxes of 5 mm on the craft's certified curve, the craft also docks without violating its barriers and without slack; its nose advances 2.4 cm further, closing the 32-mm floor gap to 10.4 mm, because both levels are computed from exact distances whereas the tabulated level absorbs the error of the rasterized training targets (Sec. VIII-B). The price is up to 1516 active corner barriers instead of one, 7.5 times the time per step, and twice the input variation; halving the boxes to 2.5 mm brings the nose 1.7 mm further (gap 8.7 mm) at 7.8 ms per step, as Proposition 5 predicts. The tabulated barrier is thus the cheaper and smoother choice where the relative pose is low-dimensional, and its conservatism is set by its training data rather than by the construction.

D. Multiple Robots

Five shapes are generated as star-shaped radial Fourier series with hull-area ratios below 0.9, scaled to radii of 0.85–1.08 m, and enclosed by certified B-spline boundaries (offsets 8–20 mm); for the ten pairs, fields with about 6-cm cells are certified in 11–16 s per pair at levels of 17–26 mm, including band and regularity. The robots are unicycles ($|v| \leq 1$ m/s, $|\omega| \leq \pi/2$ rad/s) that exchange positions on a circle of radius 4 m in a star pattern, with start and goal angles perturbed by up to 9° and 20° to avoid the central

TABLE III
Twenty Random Five-Robot Instances

	ours	closest pt.	circle
instances with a collision	0	5	0
min. true distance [mm]	+19.7	−86.8	+42.7
min. h [mm]	−2.4	−114.4	−
instances with all goals reached	13	12	20
time/step, med./max. [ms]	0.66/16.3	2.74/30.2	0.40/1.5

Horizon 15 s, step 10 ms. Negative distances: depth of the deepest vertex inside the other polygon. No step required slack.

deadlock of antipodal goals. All robots reach their goals after 10.0 s (Fig. 3); the minimum ground-truth distance is 23.0 mm, and the minimum active h is -1.1 mm, a discrete-time effect.

With the same shapes at half size, 20 random instances with poses in $[-3, 3]^2 \times \mathbb{S}^1$ compare the barriers under single-integrator dynamics (Table III). The closest-point barrier collides in 5 instances, with vertices up to 87 mm inside the other body: the closest pair of non-convex bodies jumps, the derivative is one-sided, and the constraint normal can point the wrong way just before contact. These results concern this single-closest-pair construction, the one analyzed in [10], not nonsmooth formulations over all active features. Ours and the circles never collide; the circles reach all goals, whereas ours deadlocks in 7 instances, because exact non-convex boundaries can

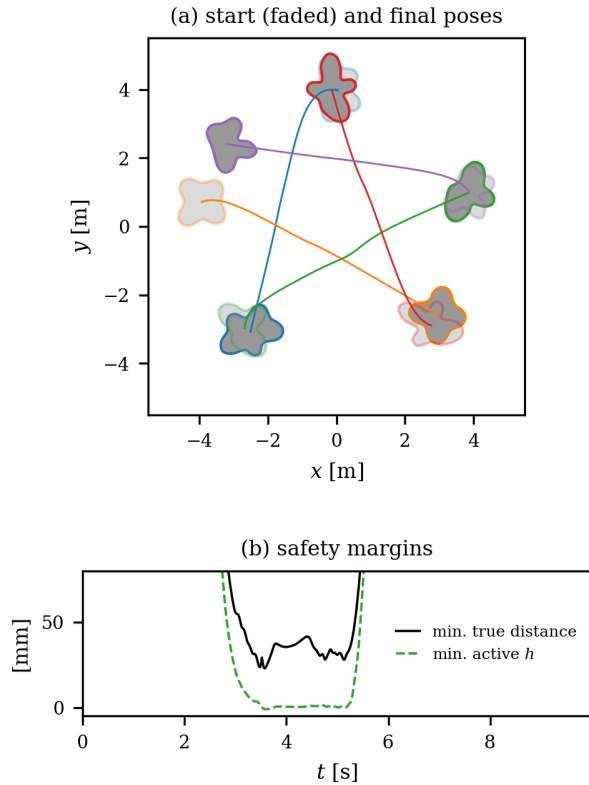


Fig. 3. Five unicycles with random non-convex shapes exchanging positions (gray: ground truth; colored: certified boundaries). (a) Start (faded) and final poses with center paths. (b) Minimum ground-truth distance over all pairs and minimum barrier value over active pairs.

hook where round surrogates slide past. On the instances completed by all methods, times and path lengths differ by less than 3%: the benefit of exact geometry appears in tight interactions such as docking. In 8 instances our barrier dips below zero, by at most 2.4 mm, when several pairs are active at full speed, while the ground-truth distance stays above 19.7 mm owing to the certified levels. In unicycle swaps of two and four robots, no instance collides (minimum distances 10.6 and 20.1 mm), the barrier’s input coefficient norm stays above 0.128 whenever $h < 5$ cm, and no step requires slack; head-on and diagonal swaps deadlock.

E. Static Obstacles

With one field for a robot among $m \in \{1, 3, 5, 7\}$ random non-convex obstacles ($122 \times 122 \times 48$ cells, 2-cm training raster, $l^* = 39\text{--}41$ mm), all runs are collision-free, and the online cost stays at 0.37–0.45 ms regardless of m , whereas the closest-point baseline needs one distance computation per nearby obstacle and grows to 2.8 ms (median) with seven.

A rocket-shaped robot (0.54 m long, nose, fuselage, two fins; 64 control points, 4-mm offset) escapes a curved maze (Fig. 4). The two wall regions are derived from a traced maze whose blind branches are removed and whose passage is smoothed; they are enclosed by nonuniform

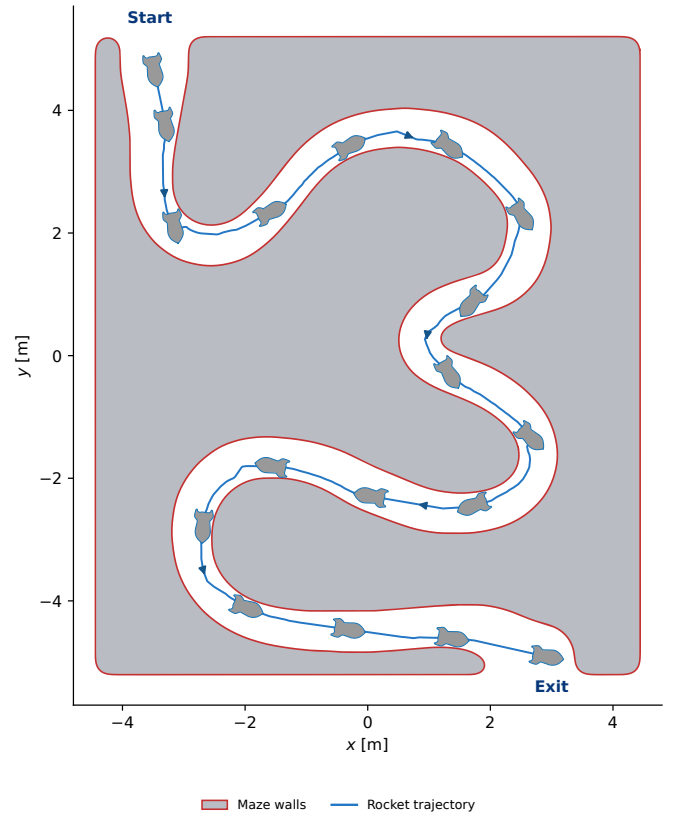


Fig. 4. Escape of a rocket-shaped robot through a curved maze with one configuration-space field and nine goal waypoints. Gray with red outlines: walls (ground truth and certified boundaries); gray with blue outlines: rocket at intermediate poses; blue: path of its reference point.

periodic B-splines (6319 and 6661 control points) with a verified maximum deviation of 1.41 mm. One $160 \times 160 \times 48$ -cell field encodes both walls ($l^* = 45.5$ mm, certified in 56 s, band and regularity verified). The nominal controller moves toward nine waypoints whose straight connections cut into the walls by up to 10 cm, so the barrier keeps the rocket off the walls: it modifies the nominal input in 75% of the 3984 steps while the rocket slides along the walls to the exit in 39.8 s. The verified clearance over every integration interval is at least 22.5 mm, the minimum barrier value is -1.1 mm, and the median online cost is 0.5 ms.

Sampling-based and ball-world baselines. On a shared fixed-orientation benchmark (a cross-shaped robot translating past a cross-shaped obstacle, six approaches, identical smooth configuration-space envelope), we compare with adaptations of the sampling-based barrier [9] and the deforming ball-world controller [15]. No method collides. Our field needs 0.106 ms per step (median) with a 29.4-mm minimum gap; the sampling barrier matches this gap only with 1024 samples, at 0.207 ms, and leaves 118.8 mm with 64; the ball-world controller needs 0.217 ms with a ten times larger input variation. These are independent adaptations on one benchmark, not reproductions.

TABLE IV
Franka Panda: 30 Trials Whose Unfiltered Motion Collides

	ours	spheres	samples	KKT
primitives	33,202	54	1975	24
certified guarantee	yes	no	no	no
trials with a collision	0	0	0	7
without margin	—	17	19	—
deepest penetration [mm]	—	10.2	13.4	29.7
certified min. distance [mm]	22.1	0.9	1.4	< 0
median of the minima [mm]	28.2	8.6	7.2	17.5
trials reaching the goal	22	20	23	19
trials needing slack	0	0	0	12
time/step, med./max. [ms]	9.4/43	1.3/10	2.3/11	46/501
barriers, max.	8210	21	188	5

Without the filter, all 30 trials collide. Primitives: boxes, spheres, surface samples, and closest-point pairs. Spheres and samples: exact obstacle SDF with a 1-cm margin unless stated; KKT: the certified fields and levels of ours. Distances: certified lower bounds between the link meshes and the obstacles. Median time over trials; Python, one CPU core.

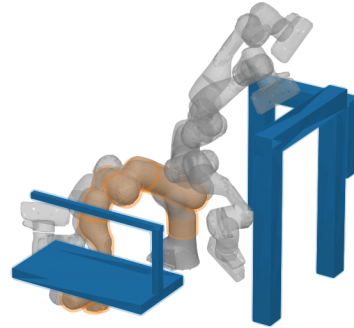
F. Franka Panda among Non-Convex Obstacles

The surface-cover barriers of Sec. VI are applied to a 7-DOF Franka Panda with the link meshes and kinematics of [11]. All eight links (the hand, without fingers, and links 1–7) are constrained against three non-convex obstacles made of rounded boxes: a gate, an inverted J, and a plate carrying a 2.4-cm rod on a leg (Fig. 5), i.e., 24 link-obstacle pairs. Each link SDF is a cubic B-spline with 12-mm cells, fitted to approximate signed distances of its mesh; its level (10.0–12.5 mm) is certified by Proposition 1 with a 4-mm grid and exact point-to-triangle distances, and its certified surface is covered by 2050–5490 boxes of side 6 mm ($r = 5.2$ mm). The obstacle fields have 2-cm cells, levels of 9.7–11.5 mm, and domains extending 15 cm beyond the obstacles. The joints are velocity-controlled ($|\dot{q}_j| \leq 1$ rad/s) with a proportional joint-space nominal controller, $\gamma = 5$, a 10-ms step, and activation threshold $\eta = 3$ cm.

Start and goal configurations are drawn uniformly within 80% of the joint ranges with the hand above 0.1 m and a clearance of at least 3 cm, and we keep the first 30 pairs whose unfiltered motion collides. Safety is evaluated without sampling: the link meshes are subdivided to edges below 4 mm (same surface), and the exact obstacle SDF at each triangle centroid minus the triangle radius lower-bounds the distance between every mesh and every obstacle. As baselines, we use the 54 collision spheres of the Franka model in [11] and about 256 surface samples per link, as in the collision-avoidance experiments of [11], each with one barrier per primitive and obstacle, the exact obstacle SDF, and a 1-cm margin.

In all 30 trials, our filtered arm is collision-free (Table IV), with a certified minimum distance of 22.1 mm between meshes and obstacles (median over trials 28.2 mm); no step requires slack. As predicted by Proposition 5, the distance at contact reflects the accumulated conservatism: the two levels (about 11 and 10 mm), the box radius, and the curvature term, compared with a single level

(a)



(b)

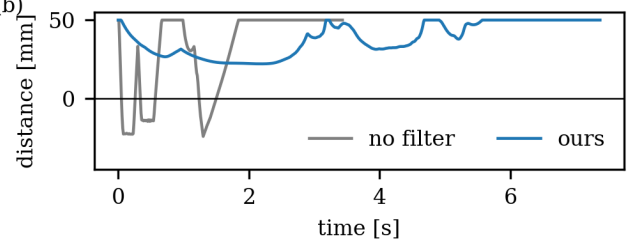


Fig. 5. Franka Panda among non-convex obstacles. (a) Poses of one trial with our filter (gray: link meshes, the pose of closest approach opaque; orange, translucent: certified link surfaces $\{\phi_A = l_A\}$ at that pose; blue: obstacles). (b) Certified lower bound of the distance between all link meshes and all obstacles, with and without the filter (values above 50 mm clipped).

of 12–20 mm in the tabulated construction. The arm reaches the goal in 22 trials and stops at $h = 0$ in front of an obstacle in the others, a deadlock of the joint-space nominal controller. The filter needs 9.4 ms per step (median over trials; maximum 43 ms) with up to 8210 active corner barriers; certified cluster pruning leaves the trajectories unchanged while reducing the cost several-fold.

With a 1-cm margin, neither baseline collides in these trials, but both come within about 1 mm of the obstacles (0.9 and 1.4 mm): the margin, not a certificate, absorbs the geometric error of the primitives. Without the margin, the spheres collide in 17 of the 30 trials, by up to 10.2 mm, because they do not enclose the meshes, and the surface samples collide in 19, by up to 13.4 mm, although every sampled barrier stays above -0.05 mm: the obstacle enters between the samples. The baselines are four to seven times cheaper per step and reach the goal about as often; what the surface cover adds is that the constraint holds on the whole continuous surface, so that no margin has to be tuned. Finally, the closest-point barrier of [10], transferred to 3D with the same certified fields and levels (one barrier per link-obstacle pair at $\mathbf{x}^* = \arg \min \phi_O$ subject to $\phi_A = l_A$, SLSQP warm-started from the previous step), collides in 7 trials by up to 29.7 mm, violates its own barrier by up to 45 mm, needs slack in 12, and costs 46 ms per

step: on non-convex links the closest point is only a local minimizer and jumps between surface patches, as in the planar comparison.

G. Two Franka Pandas in a Shared Workspace

Two Franka Pandas face each other with bases 0.9 m apart. For each of the 64 pairs of a link of arm 1 and a link of arm 2, the certified surface of the first is bounded against the SDF of the second; both links move, so each corner barrier is affine in the velocities of all 14 joints (the derivative in Sec. VI with the Jacobians of both arms), and one direction per pair suffices by Theorem 2. Starts and goals of both arms are drawn as before, with an initial clearance of 5 cm, and we keep the first 20 of 143 sampled pairs whose unfiltered motion brings the arms into contact (sampled mesh distance below 2 mm). The distance between the arms is bounded from below with exact point-to-triangle distances from the subdivided triangles of arm 1 to the meshes of arm 2. In all 20 trials the arms stay apart without slack; the barriers are active ($\min h < 5$ mm) in 11 trials, the certified minimum distance is 24.2 mm, and in 18 trials the meshes never come closer than 3 cm, where our evaluation stops resolving distances: between two links, both levels and the box radius add up (Proposition 5). Both arms reach their goals in 19 trials. The filter needs 3.2 ms per step (median over trials; maximum 57 ms) with up to 10,156 corner barriers over the 64 link pairs.

IX. Limitations

Certificates. The certificates exclude first contacts, so they protect trajectories that start collision-free, and they certify non-intersection, not a prescribed clearance (which follows by enclosing dilated ground truths). They assume exact arithmetic; our implementation uses double precision without outward rounding.

Discrete time. The guarantees hold in continuous time. With 10-ms steps, our barriers dipped to -2.4 mm in random swaps; a sampled-data tightening, e.g., with Lipschitz bounds from the Bernstein coefficients of the derivatives, is needed for the implemented controller.

Conservatism. The tabulated level is limited by the training targets; the surface-cover barriers add two levels, the box radius, and a curvature term, which is noticeable in tight spatial tasks. Finer boxes and spatially varying levels would reduce both.

Scope. Deadlocks are not resolved, and exact geometry makes them more frequent than round surrogates. The tabulated construction is limited to low-dimensional relative poses and needs one field per pair of shape types. The manipulator studies omit self-collision within an arm, the fingers, and polytopic baselines, and all experiments are in simulation.

X. Conclusion

Collision avoidance between non-convex rigid bodies reduces, through a first-contact lemma, to keeping each

body's SDF above a certified level on the boundary of the other. For planar bodies, we tabulated this condition as one smooth configuration-space field per pair and certified it with a Bernstein branch-and-bound over the explicit first-contact set; for spatial and articulated bodies, we bounded it online by a Bernstein cover of the continuous level set and corner barriers with a certified curvature majorant. Neither construction computes closest points, and both certify enclosure of the true geometry. The planar barrier never collided, unlike a closest-point barrier, and docked where bounding circles cannot; the surface-cover barriers kept a Franka Panda off three non-convex obstacles in all trials, where sphere and surface-sample barriers needed a tuned margin, and kept two Pandas apart. Sampled-data guarantees, reduced conservatism, deadlock resolution, and hardware validation are the next steps.

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